VR-Sets. Preprint.
Part VII. Truth and Self-Reference

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VII.0. Introduction

This part of the preprint considers Tarski's theorem on the undefinability of truth in the light of the operational ontology of VR-Sets. The aim is not to "resolve" or "circumvent" Tarski (this would be impossible: his theorem is a formal result, true in any sufficiently rich formal system) but to examine how the operational ontology reformulates the same situation and what subtleties this clarifies.

The part has two substantive directions:

(1) **Analysis of the liar paradox** from the standpoint of the closure principle: paradoxical self-referential statements do not specify definite operational predicates, in analogy with Russell's paradox not specifying a set (Part II §II.3).

(2) **A hierarchy of levels of truth through t**: the idea that applications of the successor operator t from VR can be interpreted as a passage to a new level of truth at which the truth of the previous level may be discussed. This accords with Tarski's approach to a hierarchy of metalanguages.

The formulation is cautious: VR-Sets *reformulates* the situation, it does not overturn Tarski's theorem. No claim of resolving classical paradoxes in the sense of "they are now gone" is being made.

VII.1. Tarski's theorem on the undefinability of truth

A brief reminder of the classical result.

Statement

Tarski's theorem (1933) asserts: for every sufficiently rich formal system T (containing arithmetic) the predicate "is true in T" is not expressible inside T. More precisely: there is no formula Tr(x) in the language of T such that for every sentence φ of the language of T:

$$T \vdash \mathrm{Tr}(\ulcorner\varphi\urcorner) \leftrightarrow \varphi$$

where $\ulcorner\varphi\urcorner$ is the encoding (Gödel number) of the sentence φ. If such a predicate were expressible, one could construct a sentence L asserting "L is false," yielding the classical liar paradox.

Significance of the result

Tarski showed that truth for the language T must be formulated in a *metalanguage* — an extension of T containing expressive means absent in T itself. This gave rise to a hierarchy of languages: object language, metalanguage, meta-metalanguage, … each with its own level of truth.

The result is universal: it applies to any formal system capable of encoding its own syntax. ZF, ZFC, PA, VR — all these systems are subject to Tarski's theorem. VR-Sets too.

VII.2. The liar paradox through the closure principle

Before discussing the hierarchy of truth, we trace how the liar paradox appears in the operational ontology. This is an exercise parallel to the dissolution of Russell's paradox in Part II §II.3.

The classical liar paradox

The sentence L = "L is false" generates a contradiction: if L is true, then L is false (by the content of L); if L is false, then L is true (since this is what it asserts). Neither truth value is consistent with L itself.

Analogy with Russell's paradox

Russell's paradox in VR-Sets is dissolved by the fact that the description R = "reveal $x \Longleftrightarrow x \notin x$" demands a contradictory response to the query "$R \in R$?" and thus does not specify an operational functionality. R *does not exist* as a set — not forbidden, but ontologically impossible.

The liar paradox has the same structure. The description of the predicate "is true" applied to L demands a contradictory response: on the query "is L true?" the procedure must return truth if and only if L is false — that is, when the procedure must return falsity. Contradictory responses. The description does not specify an operationally describable truth predicate for L.

Assertion VII.1

The paradoxical self-referential sentence L = "L is false" does not specify an operationally defined truth value. In the operational ontology L has no truth value — not because L is "neutral" or "unknowable," but because the description of truth for L is internally contradictory.

Relation to Tarski

This does not contradict Tarski's theorem. Tarski speaks of the inexpressibility of the truth predicate *for the entire language* inside the language itself. Assertion VII.1 speaks of the inexpressibility of truth *for a specific paradoxical sentence* — a particular case. VR-Sets gives an ontological interpretation: the truth predicate does not exist as an operationally describable functionality, and paradoxical sentences are the clearest witnesses to this.

For *non*-paradoxical sentences the truth predicate exists *locally*: for each specific non-self-referential φ it is operationally definable whether φ is true in a given model (for instance, by verification of the formula). What is inexpressible is the *single* truth predicate for the entire language — because the attempt to specify it runs into paradoxical sentences.

VII.3. A hierarchy of levels of truth through the successor operator t

Tarski proposed a resolution of the truth problem through a hierarchy of languages: language L_0, metalanguage L_1 (containing a truth predicate for L_0), meta-metalanguage L_2 (containing a truth predicate for L_1), and so on. Each L_{n+1} is an extension of L_n in which truth for L_n is expressible, but not for L_{n+1} itself.

In VR-Sets this hierarchy has a natural operational realisation through the successor operator t from VR.

A note on what t means here. The operator t is the same operation as in VR axiom A3 and as used in Part V to construct the von Neumann ordinals: $t(x) = x \cup \{x\}$. There is no second "t". What differs in §VII.3 is only the domain to which the operator is applied: in VR and Part V the input is $\varnothing$ and its successors O_n, generating numbers; here the input is an operational set representing an object language, generating a metalanguage. The arithmetical and the metalinguistic readings of t are two applications of the same operation to different starting objects, not two homonymous operators.

The idea

The application of t to a set A produces a new set $t(A) = A \cup \{A\}$, in which A appears as one of the elements. Substantively: t passes from A to *the level at which A may be discussed as an object*. At this new level one may formulate statements about properties of A which are inexpressible at the level of A itself.

As applied to a language: let L_n be represented by the operational set of words (sentences) of length $\leq n$ or complexity $\leq n$. Then $t(L_n) = L_n \cup \{L_n\}$ is a set containing L_n as an object. At the level of $t(L_n)$ one may speak about L_n as a whole, in particular formulate a truth predicate for the sentences of L_n.

Correspondence with Tarski

Tarski: L_0 is the object language; L_1 is the metalanguage expressing truth for L_0; and so on.

VR-Sets: L_0 is an operational set of the object language; $t(L_0)$ is the next level containing L_0 as an object; $t^2(L_0)$ is another level; and so on. The sequence of t-applications is the operational realisation of the hierarchy of metalanguages.

This is not a proof and not a new theorem. It is the indication of a structural correspondence: t in VR (succession as the passage to a new object containing the previous one as an element) corresponds structurally to the passage from a language to a metalanguage in Tarski (the metalanguage contains the object language as an object of study).

Where the correspondence is exact

Several places where the parallel operates:

(1) **The new level contains the old**: $t(A) \supset A$ in VR; the metalanguage contains the object language as a fragment.

(2) **At the new level, properties of the old are expressible**: t(A) contains A as an element, so one may formulate predicates of the form "A has property P"; in the metalanguage, truth for the object language is expressible.

(3) **The hierarchy is unbounded**: t may be applied any number of times, yielding O_n with arbitrarily large n; Tarski's hierarchy of metalanguages is likewise infinite.

(4) **Each level has its own truth predicate**: at the level $t_n(L_0)$ truth for L_0 is expressible, but not truth for $t_n(L_0)$ itself. Parallel with Tarski: each L_{n+1} expresses truth for L_n, but not for itself.

Where the correspondence is inexact

The parallel is structural, not literal. There are essential differences:

• Tarski's hierarchy is a hierarchy of *languages* including syntax and semantics. The hierarchy of t is a hierarchy of *sets*, not carrying any independent semantic load until interpreted.

• For Tarski, each L_{n+1} *contains a formal truth predicate* Tr_n for L_n. In VR-Sets the operation t merely passes to a wrapping set; the *formal expression* of the truth predicate is a separate construction that must be built at each level.

• The infinity of the t-hierarchy in VR is operational infinity (Part I §I.3). For Tarski it is ordinary infinity of ordinals; there are variants with transfinite levels. VR-Sets is restricted to ω-levels of t (by the countability of the universe — Part VI), which is weaker than the classical transfinite hierarchy.

Assertion VII.2 (cautious formulation)

The hierarchy of applications of the successor operator t from VR provides an operational realisation of the idea of Tarski's hierarchy of metalanguages, restricted to ω-levels. This is not a "resolution" of the problem of the definability of truth but an ontological interpretation of Tarski's existing resolution within a more minimalist frame.

VII.4. The duality of A1 and self-reference

In the VR system (Part I, axiom A1) the duality of implication is fixed: from F both values {F, ⊤} are reachable through →; from ⊤ — only ⊤. This asymmetry is the structural characteristic of two-valued classical logic, expressed through the primacy of F.

In VR-Numbers (Part V) the same duality motivates the two-axis structure of ℂ: the F → F axis (real) and the F → ⊤ axis (imaginary). In VR-Sets the duality of A1 provides the basis for one additional observation on self-reference.

Observation

The self-referential sentence L = "L is false" contains a *one-sided* dependence: the value of L is determined through the value of L. If one tries to represent L as an operational procedure, this procedure must first compute L in order to determine L. Circularity.

The duality of A1 gives two directions: F → F (trivial self-identity) and F → ⊤ (passage to ⊤). In the case of the self-referentiality of L, circularity stalls at F → F: the procedure refers to itself without passing to a new value.

This is not a proof; it is a speculative comment. A substantive connection between the duality of A1 and the paradoxes of self-reference is an open direction noted in Part IX as one of the promising questions for further work.

Alpha/Omega/Logos (digression)

In the working notes of the VR cycle (not included in the formal part of the preprints) a triadic personal apparatus has been explored — Alpha (inner core), Omega (reflexive observer), Logos (medium of thought). This scheme corresponds structurally to the logical triad: F (Alpha, the initial state), ⊤ (Omega, the result), → (Logos, the passage).

In the context of Part VII one may note: Tarski's hierarchy of truth is a hierarchy of *Omega*: each level of the metalanguage is the reflexive observer of the previous level. This accords with the fact that the very notion "truth for language L" is already an observation of language L from without.

This digression is not part of the formal apparatus of VR-Sets and has no theorem status. It records that the philosophical background of the VR programme naturally accords with Tarski's approach to self-reference through a hierarchy.

VII.5. Summary and transition to Part VIII

In this part:

(1) Tarski's theorem on the undefinability of truth has been recalled (§VII.1).

(2) It has been established (Assertion VII.1, §VII.2) that paradoxical self-referential sentences in the operational ontology have no truth value, for the same reason that Russell's paradox does not specify a set: the description of truth is contradictory.

(3) A parallel has been proposed between Tarski's hierarchy of metalanguages and the hierarchy of applications of the successor operator t from VR (Assertion VII.2, §VII.3). The parallel is structural, not literal; bounded to ω-levels by countability.

(4) A speculative comment on the connection of A1-duality and the paradoxes of self-reference has been fixed (§VII.4); substantive development is an open direction.

The part formulates a **cautious position**: VR-Sets does not overturn Tarski's theorem and does not resolve the paradoxes of self-reference; it gives an ontological interpretation of the existing picture within an operational frame. This is interpretation, not alternative.

What follows. Part VIII gives the final substantive discussion of the ontological position of VR-Sets. The slogan "only ∅ is, all else is doing," already used repeatedly, is unfolded into a detailed grounding of the ontological commitments. A parallel to VR-Numbers Part VI, with the accent on the specifics of a set-theoretic ontology. Briefly, also — the relation to other positions in the foundations of mathematics (Platonism, formalism, constructivism).